Name: Ritesh Ravindra Solanke

PRN: RBT24IT126

IT2262

Assignment : 2

Q1. Write the R.E. for language containing string in which every 0 is immediately followed by 11.

→ Alphabet = {0, 1}

Cond^n: whenever 0 appears, it must be followed by 11 ⇒ so 0 can appear only as 011

valid blocks : 1 & 011

∴ string is made of : (1 or 011)*

⇒ (1 + 011)*

Q2 Write R.E. for language having string which should have at least one 0 and atleast one

→ alphabets = {0, 1}

1. 0 appears before 1 somewhere in string

⇒ $(0+1)^* 0 (0+1)^* 1 (0+1)^*$

2. 1 appears before 0 somewhere in string ⇒

$(0+1)^* 1 (0+1)^* 0 (0+1)^*$

We take union

$(0+1)^* 0 (0+1)^* 1 (0+1)^* + (0+1)^* 1 (0+1)^* 0 (0+1)^*$

Expressions accepts strings like :

- 01
- 10
- 0011
- 1100
- 101010
- 01000
- 11010
- 111010

Q3. Determine R.E. over alphabet $\Sigma = (a, b)$

i) All string that contain an even no. of b's

$\Rightarrow$ R.E. $= a^*(ba^*ba^*)^*$

$a^* \Rightarrow$ any no. of a's before 1st b.

$(ba^*ba^*) \Rightarrow$ 2 b's with any no. of a's

Accepted e.g. :-

- ε (0 b's)
- aa
- bb
- abba
- aabbaa
- aaabbaaa.

ii) All strings that do not end with aa

$\Rightarrow$ End with b or

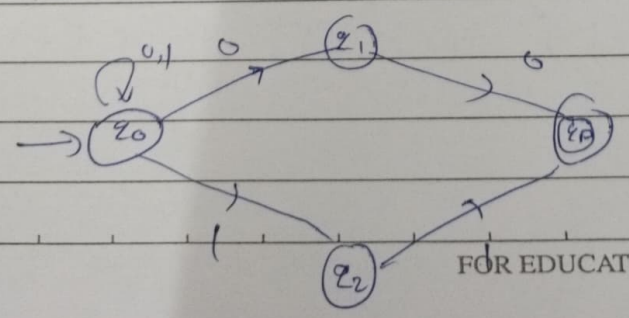
End with ab or

are of length < 2

$\Rightarrow$ R.E $\Rightarrow (a+b)^*b + (a+b)^*ab + a + b + \varepsilon$.

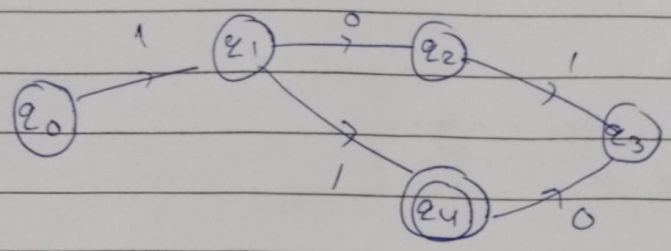
Q4. Construct finite automata for following

i) $(1+0)^*(00+11+10)$



Sundaram
FOR EDUCATIONAL USE

ii. $1(01+10)^* + 0(11+10)^*$



Q5. State and prove pumping Lemma.

→ If L is regular then $\exists$ pumping length p such that any string w in L with

$|w| \geq p$ can divided into

$w = xyz$

$|xy| \leq p$

$|y| \geq 1$

$xy^n z \in L$ for all $n \geq 0$

Application : Used to prove language aren't by Contradiction.

Q6. Describe important closure properties of R.E also write applications of R.E.

→ ① Union : $L_1 \cup L_2$

② Kleena star : L^*

③ Intersection : $L_1 \cap L_2$

④ Complement : $\bar{L}$

⑤ Difference : $L_1 - L_2$

Applications ⇒

- Compiler design
- Pattern matching

Sundaram
FOR EDUCATIONAL USE

Q7 Using pumping Lemma

$A = \{a^n b^n \mid n \geq 0\}$

→ $w = a^p b^p$

Divide : $w = xyz$

$|xy| \leq p$

$xy^0z = a^{p-k} b^p$

Now number of a ≠ number of b

∴ Not in L

Contradiction

Therefore Not Regular.